LITERATURE REVIEW

Controlling the False Discovery Rate via Knockoffs

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ANALYTICAL ABSTRACT

The paper asks how a researcher who has recorded many candidate explanatory variables for one response can select a subset while guaranteeing, in finite samples and for any correlation structure of the design, that the expected fraction of false selections stays below a chosen level. The authors construct a synthetic knockoff copy of every column of the design matrix that reproduces the original Gram matrix but is, as far as possible, uncorrelated with its twin. A statistic that compares each original variable with its knockoff, here the order of entry on the Lasso path, has symmetric sign for null variables, which allows the number of false discoveries above any threshold to be estimated by the count of negative statistics. A data-dependent threshold based on that ratio yields exact false discovery rate control for the knockoff+ variant and control of a modified rate for the basic variant. The theory requires Gaussian noise, a fixed design and at least as many observations as variables. Simulations with 3000 observations and 1000 variables show about 61 percent power against 49 percent for the Benjamini and Hochberg procedure at a 20 percent target, and an application to HIV drug resistance gives broadly comparable selections. The main limitation is that power is shown only empirically and the guarantee excludes the high-dimensional regime.

1 Introduction

1.1 Paper identification and classification

The paper is a methodological contribution in theoretical and applied statistics, published in The Annals of Statistics in 2015 and first circulated as arXiv 1404.5609. The review is based on the full text of the final arXiv version (version 3, stat.ME, 14 October 2015, 33 pages), including all sections, the simulations in Figures 1 to 9, Table 1, the HIV data application and the reference list. The supplementary material, which contains the proofs of the theorem on sequential testing and of the binomial inequality, was not available in this run, so those proofs are reconstructed or flagged below. The field is statistical inference; the subfield is multiple hypothesis testing and post-selection error control for sparse regression. The paper is classified under Applied Statistics because its object is a usable procedure for scientific discovery problems such as genomics, although its core is a finite-sample theorem.

1.2 Executive understanding

WHAT IS THE PAPER TRYING TO DISCOVER?

It asks whether a variable selection rule exists that controls the expected proportion of false discoveries exactly, at finite sample size, without knowing the noise level, the number of true variables or the amplitude of the coefficients, and without assuming the features are independent. The answer given is yes for the Gaussian linear model with a fixed design when the number of observations is at least the number of variables.

WHY DOES THE QUESTION MATTER?

Scientists routinely regress one outcome on hundreds or thousands of recorded features and report the selected ones as discoveries. If a large share of those discoveries are spurious, follow-up experiments fail to replicate. The false discovery rate gives a quantity that has a direct operational reading: at a 10 percent level, about nine of every ten reported variables are real. Classical tools that deliver this guarantee assume independence or positive dependence that regression z-scores rarely satisfy.

WHAT WAS KNOWN BEFORE, AND WHAT IS MISSING?

Benjamini and Hochberg (1995) control the rate under independence, Benjamini and Yekutieli (2001) under arbitrary dependence at a cost of a logarithmic factor, and stability selection (Meinshausen and Buhlmann, 2010) offers asymptotic selection consistency but no finite-sample false discovery bound in general. Permutation heuristics are widely used but fail when features are correlated and some signals exist. What was missing is a procedure that exploits a powerful selection statistic such as the Lasso while retaining an exact finite-sample guarantee under arbitrary design correlation.

WHAT DOES THE PAPER PROPOSE, HOW DOES IT INVESTIGATE, AND WHAT DOES IT FIND?

It proposes the knockoff filter: build a design copy that mimics the original correlations, compute antisymmetric statistics that compare each variable to its copy, and choose a threshold from the ratio of negative to positive statistics. It proves exact control for the knockoff+ version by viewing the procedure as a stopping time of a supermartingale in a sequential testing problem. It evaluates the method against permutation knockoffs, against Benjamini and Hochberg variants and in an HIV application. It finds control at or below the nominal level in nearly all designs and substantially higher power than the valid competitors, with a single failure for the basic version at a feature correlation of 0.9.

2 Research Problem and Hypothesis

2.1 Research question

PAPER Can a variable selection procedure for the model $y = X\beta + z$ with $z \sim N(0, \sigma^2 I)$, fixed $X \in \mathbb{R}^{n \times p}$ and $n \geq p$, be constructed such that the expected fraction of selected variables with $\beta_j = 0$ is at most q for every β and every σ, regardless of the correlations among the columns of X, while retaining enough power to find weak signals?

2.2 Hypotheses

- Null features can be made statistically indistinguishable from their knockoffs, so the sign of the comparison statistic for each null is a fair coin, independent of everything else.
- The count of negative statistics beyond a threshold therefore estimates, conservatively, the count of false positive statistics beyond it, which supports a data-driven threshold with provable control.
- A statistic that is powerful for sparse regression, here the Lasso entry time, retains power after being converted to a one-bit comparison, and outperforms Benjamini and Hochberg procedures applied to least-squares z-scores when the proportion of nulls is high.
- Permutation-based surrogates for the knockoffs do not deliver the same exchangeability and therefore lose control under correlation.

2.3 Existing literature and research gap

PAPER The authors position the work against Benjamini and Hochberg (1995), Benjamini and Yekutieli (2001), Storey (2002), permutation methods, Miller's dummy variables and rotations (Miller, 1984; 2002), the sequential Lasso procedures of G'Sell, Wager, Chouldechova and Tibshirani (2013, later 2016), stability selection and the emerging high-dimensional confidence interval literature (Javanmard and Montanari, 2014; van de Geer et al., 2014; Zhang and Zhang, 2014). CRITIQUE The gap is real and precisely stated: no earlier method combined finite-sample exactness, arbitrary design,

unknown noise level and use of a modern selection statistic. The paper is somewhat narrower than its introduction suggests, because the cases where finite-sample claims were most needed, $p > n$, are explicitly deferred.

2.4 Claimed contribution

CRITIQUE The paper claims a new construction (knockoff features), a new class of statistics (antisymmetric and sufficient), exact FDR control for knockoff+ and modified-FDR control for the basic version, an equivalence with a sequential testing procedure, and empirical superiority in power. The theoretical claims are demonstrated by proof (the proof of the main step is in the supplement, which was not reviewed here). The power claim is demonstrated only by simulation and one application without ground truth. The claim that the method works for ”any design” is correct only conditional on $n \geq p$ and on invertibility of $X^\top X$.

3 Theoretical Framework and Mechanism

3.1 Model and error criterion

PAPER The data obey the linear Gaussian model and the target is the false discovery rate of the selected set $\hat{S}$:

$$y = X\beta + z, \qquad z \sim N(0, \sigma^2 I_n), \tag{1}$$

$$\mathrm{FDR} = \mathbb{E}\left[\frac{\left|\{j : \beta_j = 0,\ j \in \hat{S}\}\right|}{\left|\{j : j \in \hat{S}\}\right| \vee 1}\right]. \tag{2}$$

$y \in \mathbb{R}^n$	Response vector.
$X \in \mathbb{R}^{n \times p}$	Known design, columns normalised so that $\Sigma_{jj} = \|X_j\|_2^2 = 1$.
$\beta \in \mathbb{R}^p$	Unknown coefficients; a null variable has $\beta_j = 0$.
$\hat{S}$	Selected set; the denominator is set to one when nothing is selected, so the fraction is zero in that case.

Equation (2) averages the false discovery proportion over the Gaussian noise. A procedure controls FDR at level q if the bound holds for every β. The rate is a property of the expectation, not of a single data set, so a given realisation can have a larger proportion of false selections.

3.2 Constructing the knockoff design

PAPER Let $\Sigma = X^\top X$, assumed invertible. The knockoff matrix $\tilde{X} \in \mathbb{R}^{n\times p}$ is required to satisfy

$$\tilde{X}^\top \tilde{X} = \Sigma, \qquad X^\top \tilde{X} = \Sigma - \operatorname{diag}\{s\}, \tag{3}$$

which is the statement that the augmented matrix $[X\,\tilde{X}]$ has Gram matrix

$$G = \begin{pmatrix} \Sigma & \Sigma - \operatorname{diag}\{s\} \\ \Sigma - \operatorname{diag}\{s\} & \Sigma \end{pmatrix}. \tag{4}$$

$s \in \mathbb{R}^p_+$ Free nonnegative vector; $1 - s_j = X_j^\top \tilde{X}_j$ is the correlation between a variable and its own knockoff.

$\Sigma - \operatorname{diag}\{s\}$ Equals Σ off the diagonal, so $X_j^\top \tilde{X}_k = X_j^\top X_k$ for all $j \neq k$: a knockoff correlates with every other original exactly as the original does.

A matrix $\tilde{X}$ with Gram G exists if and only if $G \succeq 0$. By a Schur complement argument the authors state that this holds if and only if $\operatorname{diag}\{s\} \succeq 0$ and $2\Sigma \succeq \operatorname{diag}\{s\}$. The reviewer verifies this step: the Schur complement of the upper-left block is $\Sigma - (\Sigma - S)\Sigma^{-1}(\Sigma - S) = 2S - S\Sigma^{-1}S$ with $S = \operatorname{diag}\{s\}$, and this equals $S^{1/2}\big(2I - S^{1/2}\Sigma^{-1}S^{1/2}\big)S^{1/2}$, which is positive semidefinite if and only if $S^{1/2}\Sigma^{-1}S^{1/2} \preceq 2I$, equivalent to $S \preceq 2\Sigma$.

When $n \geq 2p$ the construction is explicit:

$$\tilde{X} = X\big(I - \Sigma^{-1}S\big) + \tilde{U}C, \qquad C^\top C = 2S - S\Sigma^{-1}S, \tag{5}$$

where $\tilde{U} \in \mathbb{R}^{n\times p}$ has orthonormal columns orthogonal to the column space of X ($\tilde{U}^\top X = 0$, possible because $n \geq 2p$) and C is a Cholesky factor. The reviewer checks that this satisfies (3): since $\tilde{U}^\top X = 0$ and $\tilde{U}^\top \tilde{U} = I$,

$$\tilde{X}^\top \tilde{X} = (I - \Sigma^{-1}S)^\top \Sigma (I - \Sigma^{-1}S) + C^\top C = \Sigma - 2S + S\Sigma^{-1}S + 2S - S\Sigma^{-1}S = \Sigma,$$

$$X^\top \tilde{X} = \Sigma(I - \Sigma^{-1}S) = \Sigma - S.$$

The two requirements are met. Economic or scientific meaning is replaced here by a geometric one: each knockoff is a vector that sits in the same position relative to the other original features as its twin, but is pushed out of the span of X by the amount $1 - s_j$ so that it carries no true signal.

3.3 Choice of the vector s

PAPER Power requires that a signal variable enters a selection path before its knockoff, which is more likely when $\tilde{X}_j$ is nearly orthogonal to X_j and hence to the signal. Two choices are proposed:

$$s_j = 2\lambda_{\min}(\Sigma) \wedge 1 \quad \text{(equi-correlated)}, \tag{6}$$

$$\min_{s} \sum_{j} (1 - s_j) \quad \text{subject to} \quad 0 \le s_j \le 1,\ \operatorname{diag}\{s\} \preceq 2\Sigma \quad \text{(SDP)}. \tag{7}$$

The equi-correlated choice makes every $\langle X_j, \tilde{X}_j \rangle = 1 - (2\lambda_{\min}(\Sigma) \wedge 1)$ and minimises its absolute value among equal-variance constructions. The semidefinite programme minimises the average original-knockoff correlation. INFERENCE The equi-correlated choice is limited by the smallest eigenvalue of Σ, so a single nearly collinear block of the design degrades the distinguishability of every variable (see Figure 3 of this review and Section 8).

3.4 Statistics, the Lasso path and the threshold

PAPER The statistic must satisfy sufficiency ($W = f([X\,\tilde{X}]^\top [X\,\tilde{X}], [X\,\tilde{X}]^\top y)$) and antisymmetry (swapping X_j with $\tilde{X}_j$ flips the sign of W_j). The leading example runs the Lasso on the augmented design,

$$\hat{\beta}(\lambda) = \arg\min_{b} \ \tfrac{1}{2}\|y - [X\,\tilde{X}]\, b\|_2^2 + \lambda \|b\|_1, \qquad Z_j = \sup\{\lambda : \hat{\beta}_j(\lambda) \neq 0\}, \tag{8}$$

$$W_j = (Z_j \vee \tilde{Z}_j) \cdot \begin{cases} +1, & Z_j > \tilde{Z}_j \\ -1, & Z_j < \tilde{Z}_j \end{cases} \tag{9}$$

- Z_j — Largest penalty at which the original variable j is in the Lasso model: the entry time on the path.
- $\tilde{Z}_j$ — The same quantity for the knockoff, written Z_{j+p} in the general notation.
- W_j — Magnitude is the later of the two entry levels; sign is positive if the original enters first. Large positive values are evidence against $\beta_j = 0$.

Given a target level q, the data-dependent thresholds are

$$T = \min\left\{ t \in \mathcal{W} : \ \frac{\left|\{j : W_j \le -t\}\right|}{\left|\{j : W_j \ge t\}\right| \vee 1} \le q \right\},$$
$$T^{+} = \min\left\{ t \in \mathcal{W} : \ \frac{1 + \left|\{j : W_j \le -t\}\right|}{\left|\{j : W_j \ge t\}\right| \vee 1} \le q \right\}, \tag{10}$$

with $\mathcal{W} = \{|W_j|\} \setminus \{0\}$, $T = +\infty$ if the set is empty, and $\hat{S} = \{j : W_j \ge T\}$. The first is the knockoff filter and the second the knockoff+ filter. The ratio in (10) is the knockoff estimate of the false discovery proportion.

3.5 Why the estimator of the false discovery proportion works

PAPER The key property is that the signs of the null statistics are i.i.d. fair coins, independent of the magnitudes. The authors state it as a lemma.

> **LEMMA 1 (I.I.D. SIGNS FOR THE NULLS)**
>
> Let $\varepsilon \in \{\pm 1\}^p$ be independent of W, with $\varepsilon_j = +1$ for non-null j and $\varepsilon_j \overset{\text{i.i.d.}}{\sim} \{\pm 1\}$ for null j. Then $(W_1, \dots, W_p) \overset{d}{=} (W_1\varepsilon_1, \dots, W_p\varepsilon_p)$.

The proof rests on two exchangeability facts. First, by construction (4), swapping X_j and $\tilde{X}_j$ for any set S does not change the Gram matrix. Second, for a set S of nulls, $[X\,\tilde{X}]_{\text{swap}(S)}^\top y \overset{d}{=} [X\,\tilde{X}]^\top y$, because $y \sim N(X\beta, \sigma^2 I)$ and the swapped vector has the same covariance (the Gram is unchanged) and the same mean (since $X_j^\top X\beta = \tilde{X}_j^\top X\beta$ when $\beta_j = 0$ and $X_j^\top X_k = \tilde{X}_j^\top X_k$ for $k \neq j$). Antisymmetry then gives $W_{\text{swap}(S)} = (W_1\varepsilon_1, \dots)$ with $\varepsilon_j = -1$ on S, and sufficiency lets one write $W_{\text{swap}(S)} = f(\cdot) \overset{d}{=} f(\cdot) = W$. Consequently, for any t, $\left|\{j \text{ null} : W_j \le -t\}\right| \overset{d}{=} \left|\{j \text{ null} : W_j \ge t\}\right|$, and conditional on $|W|$ both are Binomial with the same parameters. This supports the inequality

$$\text{FDP}(t) = \frac{\left|\{j : \beta_j = 0, W_j \ge t\}\right|}{\left|\{j : W_j \ge t\}\right| \vee 1} \approx \frac{\left|\{j : \beta_j = 0, W_j \le -t\}\right|}{\left|\{j : W_j \ge t\}\right| \vee 1} \le \frac{\left|\{j : W_j \le -t\}\right|}{\left|\{j : W_j \ge t\}\right| \vee 1} = \widehat{\text{FDP}}(t). \tag{11}$$

The inequality is conservative because negative statistics from non-nulls are also counted. It is tight when most signals enter before their knockoffs.

3.6 From the estimator to a theorem

PAPER Write $V^{\pm}(t) = \left|\{j \text{ null} : |W_j| \ge t,\ \text{sign}(W_j) = \pm 1\}\right|$. For knockoff+ the proof sketch in the paper proceeds in three steps. The first bounds the realised proportion at the threshold:

$$\text{FDP} \le \frac{1 + \left|\{j : W_j \le -T\}\right|}{\left|\{j : W_j \ge T\}\right| \vee 1} \cdot \frac{\left|\{j : \beta_j = 0, W_j \ge T\}\right|}{1 + \left|\{j : \beta_j = 0, W_j \le -T\}\right|} \le q \cdot \frac{V^+(T)}{1 + V^-(T)}, \tag{12}$$

the first inequality because the null negatives are a subset of all negatives, the second by the definition of T. The second step shows that T is a stopping time and that $V^+(t)/(1 + V^-(t))$ is a supermartingale in the filtration that reveals $|W|$ and counts down in t, so optional stopping gives $\mathbb{E}[V^+(T)/(1+V^-(T))] \le \mathbb{E}[V^+(0)/(1+V^-(0))]$. The third step uses $V^+(0) \sim \text{Binomial}(p_0, 1/2)$ and $V^-(0) = p_0 - V^+(0)$, where p_0 is the number of nulls. The paper defers the inequality $\mathbb{E}[V^+(0)/(1+p_0-V^+(0))] \le 1$ to the supplement. The reviewer derives it directly. For $B \sim \text{Binomial}(p_0, 1/2)$ and $b \ge 1$, $\binom{p_0}{b}\frac{b}{p_0-b+1} = \binom{p_0}{b-1}$, hence

$$\mathbb{E}\left[\frac{B}{1 + p_0 - B}\right] = 2^{-p_0} \sum_{b=1}^{p_0} \binom{p_0}{b-1} = 2^{-p_0}\left(2^{p_0} - 1\right) = 1 - 2^{-p_0} < 1. \tag{13}$$

Combining (12) and (13) yields FDR $\leq q$ for knockoff+. The basic filter has the numerator 0 in place of 1, which leaves a residual that is controlled by the modified rate of Theorem 1.

> **THEOREM 1 AND THEOREM 2**
>
> For any $q \in [0,1]$, the knockoff filter satisfies $\mathbb{E}\left[\frac{\left|\{j : \beta_j = 0,\, j \in \hat{S}\}\right|}{\left|\{j : j \in \hat{S}\}\right| + q^{-1} - 1}\right] \leq q$, and the knockoff+ filter satisfies FDR $\leq q$. The expectation is over the Gaussian noise with X and $\tilde{X}$ fixed.

3.7 Extension to p smaller than n smaller than 2p

PAPER If $n < 2p$ the orthogonal complement is too small to build $\tilde{U}$. The authors append $2p - n$ rows of zeros to X, draw $y' \sim N(0, \hat{\sigma}^2 I_{2p-n})$ and append it to y, where $\hat{\sigma}^2$ comes from the residual sum of squares $\|y - X\hat{\beta}^{LS}\|_2^2 \sim \sigma^2 \chi^2_{n-p}$. The augmented problem has $2p$ rows. **CRITIQUE** Control is then approximate, since $\hat{\sigma}$ replaces σ, and the output depends on an extra random draw. The paper does not bound the resulting loss of control or quantify its dependence on $n - p$.

3.8 Equivalence with sequential testing

PAPER Section 5 reframes the procedure as a sequential testing rule on ordered one-bit p-values. After sorting $|W_1| \geq |W_2| \geq \ldots \geq |W_m| > 0$, set

$$p_j = \begin{cases} 1/2, & W_j > 0 \\ 1, & W_j < 0 \end{cases} \tag{14}$$

which are valid super-uniform p-values for nulls by Lemma 1. Selective SeqStep with $c = 1/2$ rejects H_j for $j \leq \hat{k}$ with $p_j \leq c$, where $\hat{k}$ is the largest k with $\frac{\{0 \text{ or } 1\} + \left|\{j \leq k : p_j > c\}\right|}{\left|\{j \leq k : p_j \leq c\}\right| \vee 1} \leq \frac{1-c}{c} q$. With $c = 1/2$ the factor $(1-c)/c$ equals one and the expression coincides with (10). Theorem 3 then gives the FDR guarantees under independent null p-values and arbitrary ordering that does not depend on the p-values.

4 Data and Empirical Strategy

4.1 Data

PAPER The paper has two empirical components. The simulation uses a design $X \in \mathbb{R}^{3000\times 1000}$ with i.i.d. $N(0,1)$ entries, columns normalised, $k = 30$ non-null coefficients with random signs and amplitude $A = 3.5$, and $y \sim N(X\beta, I)$. The paper states that 3.5 is approximately the expected value of $\max_j |X_j^\top z|$, so signals sit near the largest noise correlation. The real-data component uses the HIV-1 drug-resistance data of Rhee et al. (2006), with the response equal to the log-fold increase in laboratory-tested resistance and binary design entries indicating the presence of each mutation. Mutations appearing fewer than three times are dropped and duplicated columns are removed. Validation uses treatment-selected mutation panels of Rhee et al. (2005).

Table 1. **HIV drug-resistance data sets used in the application**

Drug class	Drugs	Samples	Positions genotyped	Mutations seen at least 3 times
Protease inhibitors (PI)	7	846	99	209
Nucleoside RT inhibitors (NRTI)	6	634	240	287
Nonnucleoside RT inhibitors (NNRTI)	3	745	240	319

Source: copied from Section 4 of the paper (official). Sample sizes and variable counts are slightly smaller per drug because of missing resistance measurements; the paper gives the per-drug values only in Figures 7 to 9.

4.2 Empirical specification and identification

EVIDENCE There is no causal identification problem in the usual econometric sense: the target is the support of β in a correctly specified linear model, and randomness comes only from the noise z with X fixed. The identifying variation for the guarantee is the exchangeability of null variables and their knockoffs under the swap operation, not any exogenous shock. The analysis treats different mutations at the same position as distinct features and assumes an additive model with no interactions. Validation compares selected positions rather than mutations with the treatment panels. The panels are derived from the comparison of treated and untreated patients, which is a different type of evidence, so agreement is treated as replication.

4.3 Data limitations and biases

- The treatment-selected mutation panel is an approximation of the ground truth, not a gold standard, so unmatched selections are not necessarily false discoveries and matched ones are not necessarily true effects.
- Simulations generate the design with i.i.d. Gaussian rows, which is the benign case for the Lasso and for the semidefinite programme. The AR(1) correlation experiment is a single family of dependence structures.

- The amplitude $A = 3.5$ was chosen to be near the maximal noise correlation, which maximises the discriminating power of the experiment but also fixes one regime. The relative standing of the methods outside it is shown only through the amplitude sweep in Figure 4 of the paper.
- Biological data are observational samples of viral isolates from patients, subject to selection by prior treatment and to shared ancestry among isolates, which violates independent noise.
- Mutations appearing fewer than three times are dropped, a data-dependent filter based on the design only, which is allowed with fixed-design theory but removes rare resistance mutations.

5 Results

5.1 False discovery rate and power against Benjamini and Hochberg variants

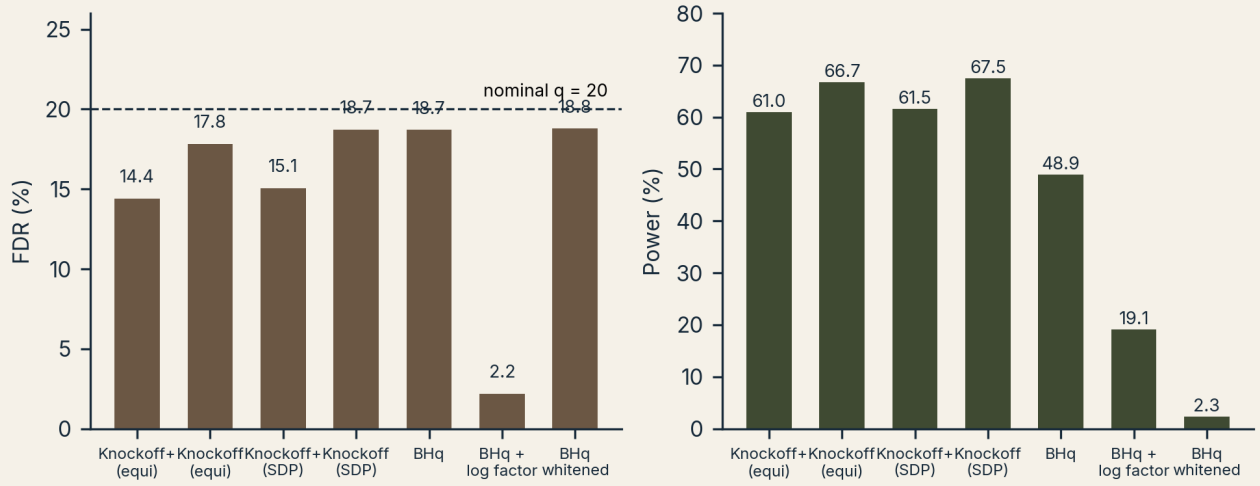
EVIDENCE In the main simulation (3000 observations, 1000 variables, 30 signals, amplitude 3.5, nominal level 20 percent, average over 600 trials), all seven methods have empirical FDR at or below the target, and the knockoff methods have the highest power.

Table 2. **Table 1 of the paper: FDR and power, n = 3000, p = 1000, k = 30, A = 3.5, nominal q = 20 percent**

Method	FDR (%)	Power (%)	Theoretical guarantee
Knockoff+ (equi-correlated)	14.40	60.99	Yes
Knockoff (equi-correlated)	17.82	66.73	No (modified FDR)
Knockoff+ (SDP)	15.05	61.54	Yes
Knockoff (SDP)	18.72	67.50	No (modified FDR)
Benjamini and Hochberg (BHq)	18.70	48.88	No
BHq with log-factor correction	2.20	19.09	Yes
BHq with whitened noise	18.79	2.33	Yes

Source: copied from Table 1 of Barber and Candès (2015) (official).

Figure 1. **FDR and power of the seven selection rules in the main simulation (nominal level 20 percent)**



Source: values copied from Table 1 of the paper (official); chart drawn by the reviewer. Percent; n = 3000, p = 1000, k = 30, A = 3.5, 600 trials.

EVIDENCE The table answers one question: among the procedures with a theoretical guarantee, how much power is lost? The two procedures with guarantees that are not knockoffs have very low power: the log-factor correction divides q by $S(p) = \sum_{i \le p} 1/i \approx \log p + 0.577 \approx 7.5$ for $p = 1000$, so it works at an effective level of about 2.7 percent, and the whitening procedure adds noise of variance $\sigma^2(\lambda_0^{-1} I - \Sigma^{-1})$ which destroys most of the signal. Knockoff+ achieves about 61 percent power with realised FDR near 14 to 15 percent, a margin of about 5 points below the target. The basic filter reaches 67 percent power at FDR between 18 and 19 percent. In intuitive units, among 30 true variables, knockoff+ finds about 18 and the Benjamini and Hochberg rule about 15. The result does not establish that knockoffs dominate BHq: BHq with ordinary least-squares z-scores has no guarantee and attains an FDR of 18.7 percent here, so the comparison of 61 versus 49 percent is between a guaranteed and an unguaranteed procedure that also happens to control the rate in this design.

5.2 Sensitivity to sparsity, amplitude and correlation

EVIDENCE Figures 3 to 5 of the paper vary one parameter at a time (sparsity $k = 10, \dots, 200$, amplitude $A = 2.8, \dots, 4.2$, AR(1) correlation $\rho = 0, \dots, 0.9$, with correlation $\rho^{|j-k|}$ between columns). All three methods hold FDR near or below 20 percent with one exception: the basic knockoff filter reaches 26.67 percent at $\rho = 0.9$, where fewer than five variables are typically selected. The paper links this directly to Theorem 1, since with so few discoveries the extra $q^{-1} - 1 = 4$ in the denominator of the modified rate is large relative to the number selected. Power of the knockoff procedures is at least that of BHq in all settings and is far higher at low and moderate k, with BHq catching up as k grows and the problem becomes dense. All methods lose power with high correlation and low amplitude.

5.3 Orthogonal designs

EVIDENCE With an orthogonal 2000×1000 design, 200 signals ($\pi_0 = 0.8$), $q = 0.2$ and 1000 trials (Figure 6 of the paper), knockoff+ and BHq have almost identical power. The FDR of BHq equals $\pi_0 q = 0.16$ at every amplitude, as proved by Benjamini and Hochberg (1995). The FDR of knockoff+ approaches $q = 0.20$ when power is high, so the procedure implicitly adapts to the proportion of nulls, and is far below q when power is near zero. INFERENCE The BHq conservatism of π_0 (here a 20 percent relative loss) is recovered because the negative-W count estimates the null count directly rather than assuming $\pi_0 = 1$.

5.4 Comparison with permutation knockoffs

EVIDENCE With $X \in \mathbb{R}^{300 \times 100}$ drawn from $N(0, \Theta)$ with $\Theta_{ij} = 0.3$, $y = 3.5(X_1 + \cdots + X_{30}) + z$ and target 20 percent (1000 trials), the knockoff filter attains an FDR of 12.29 percent and the permutation variant 45.61 percent (Figure 2 of the paper). The mechanism is visible in the Lasso path: only the original nulls are correlated with the signals, so they enter early, whereas permuted copies enter at very small λ, which makes the negative-statistic count a gross under-estimate of false positives. The result establishes that exchangeability, not merely preserved marginal Gram structure, is required.

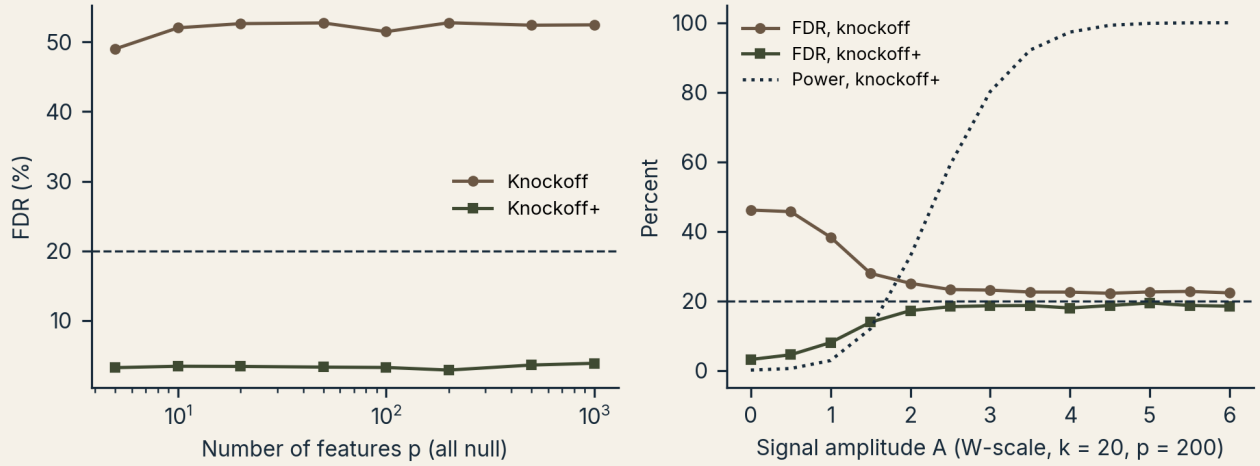
5.5 HIV drug resistance

EVIDENCE For each of 16 drugs the response is regressed on mutation indicators with $q = 20$ percent for knockoff and BHq (Figures 7 to 9). Selected positions are compared with the treatment-selected mutation panels, and the paper reports that both methods agree well with the panels and that knockoffs are slightly better overall, with variability across drugs. One drug lies in the regime $p < n < 2p$ and uses the row-augmentation of Section 2.1.2. No FDR or power can be computed in this application because the truth is unknown. The result shows plausibility and replication, not the property of FDR control.

5.6 Non-Gaussian noise with a sparse real design

EVIDENCE Using the actual NNRTI design (747×319, binary and sparse), $k = 20$ random signals drawn from $3.5 \cdot N(0, 1)$, and noise resampled from the empirical residual distribution (excess kurtosis 9.24, equivalent to a t distribution with 4.65 degrees of freedom), 1000 trials give FDR of 25.72 percent (knockoff), 20.31 percent (knockoff+) and 25.47 percent (BHq), and power of 63.50, 60.67 and 69.42 percent. The reviewer notes that the knockoff+ FDR of 20.31 percent exceeds the nominal 20 percent by 0.31 points, so the guarantee is approximately but not exactly retained under heavy tails, and the statement that robustness is verified is supported only to a modest tolerance. In this experiment BHq has the highest power, which weakens a blanket claim of knockoff superiority.

Figure 2. **Reviewer simulation on the W-statistic layer: false discovery rate of the knockoff and knockoff+ thresholds**



Source: generated by the reviewer, not from the paper. Null W-statistics are random sign times the absolute value of a standard normal; signals have W equal to A plus a standard normal. Nominal q = 20 percent; 4000 trials (left) and 1500 trials (right); p = 200 and k = 20 in the right panel. This isolates the threshold rule and does not run the Lasso.

5.7 Reviewer simulation: why the plus matters

EVIDENCE A reviewer-run simulation applies the threshold rules (10) directly to synthetic W-statistics (Figure 2 of this review, generated). Under the global null, the basic knockoff filter has FDR of about 49 to 53 percent for p between 5 and 1000, while knockoff+ has FDR between about 3 and 4 percent. The basic filter is therefore far from controlling the FDR in the global-null case, although it satisfies its modified-rate theorem, since with R selected the denominator in Theorem 1 is $R + 4$. With 20 signals among 200 variables the basic filter has FDR of 46 percent at amplitude $A = 0$, falling to about 22.5 percent for $A \geq 3.5$, which is slightly above the nominal level even at high power. Knockoff+ stays near 17 to 19 percent for $A \geq 2$. Power of the two is within about 2 points from $A = 3.5$ upward. INFERENCE These numbers are produced by a stylised model and must not be read as properties of the Lasso statistic, but they quantify the paper's remark that the added one in knockoff+ is necessary when few discoveries are made.

6 Robustness and Additional Tests

Table 3. **Robustness tests: concern, test, result and assessment**

Concern	Test reported	Result	Does it resolve the concern?
Is the permutation shortcut as good as the construction?	Permutation copies in place of knockoffs, equicorrelation 0.3 (Figure 2 of the paper)	FDR 45.61 percent against 12.29 percent	Yes for the narrow claim: permutation is invalid under correlation. Not a test of other cheap surrogates.

Concern	Test reported	Result	Does it resolve the concern?
Does control hold at high correlation?	AR(1) correlation 0 to 0.9 (Figure 5 of the paper)	Knockoff reaches 26.67 percent at 0.9; knockoff+ holds	Partly: confirms the theory for the plus version, but shows the basic filter is unreliable with few discoveries.
Is the result special to one sparsity or signal size?	Sweeps in k and A (Figures 3 and 4 of the paper)	Control is maintained; power advantage largest for sparse models	Partly: one design family, one n over p ratio of 3.
Is the result sensitive to the construction of s?	Equi-correlated against SDP (Table 1)	SDP gives slightly higher power (61.54 against 60.99)	Yes within this design; the difference is small and no standard errors are reported.
Do the guarantees survive non-Gaussian noise?	Empirical noise with excess kurtosis 9.24 on a sparse real design	Knockoff+ 20.31 percent, knockoff 25.72 percent	Partly: approximate, not exact, with one design and one noise law.
Is the real-data agreement informative?	Selected positions compared with TSM panels (Figures 7 to 9 of the paper)	Knockoff slightly better than BHq, with variability	Weakly: the panel is an approximation, no uncertainty is quantified, and with a single run per drug the difference is not tested.
Does the p between n and 2p extension work?	Used on one HIV drug	Reported but not separately evaluated	No: no simulation of control under the augmentation is reported.

Source: summarised by the reviewer from Sections 3 and 4 of the paper (paper results) with the reviewer's assessment (critique).

6.1 Reviewer assessment

CRITIQUE The robustness evidence is adequate for the statement that the theory is not vacuous, and the finding for the plus version is consistent with the proof. It is thin for the statement that the method is practically robust: no study varies the ratio n/p, none reports Monte Carlo standard errors (only means over 200 to 1000 trials), and the misspecification experiment uses only one design. The power comparisons were not repeated over alternative selection statistics, although the framework advertises that many are possible.

7 Critical Evaluation

7.1 What the paper establishes

ESTABLISHED

Under Gaussian noise, a fixed full-rank design and $n \geq 2p$, the knockoff+ filter has FDR $\leq q$ for every coefficient vector, and the basic filter controls the modified rate. Permutation copies are not valid substitutes under correlated designs. Knockoff+ retains substantially higher power than the valid Benjamini and Hochberg variants in the reported simulations.

SUPPORTED BUT CONDITIONAL

The extension to $p < n < 2p$ through row augmentation with an estimated noise level (approximate, and relies on a large $n - p$). The statement that knockoffs outperform BHq in power in sparse regimes (true in the reported designs; BHq with OLS z-scores lacks a guarantee, and in the heavy-tailed experiment BHq has higher power).

SUGGESTIVE

That the selections in the HIV data correspond to real, replicable effects (agreement with panels that are themselves approximate). That the procedure is robust to non-Gaussian noise (one experiment).

NOT ESTABLISHED

Any theoretical statement about power or false negative rate. Validity when $p > n$ or when $X^\top X$ is singular. Validity under random designs, heteroskedastic or dependent errors, or model misspecification. That the one-bit comparison is the best way to use the information in W (the authors suggest multiple knockoffs, but do not test it).

7.2 Strengths

STRENGTHS RECOGNISED BY THE AUTHORS

- Exact finite-sample control for any fixed design with $n \geq p$ and unknown noise level, which removes the need for asymptotics or tuning to a sparsity level.
- Flexibility in the choice of W: any statistic obeying sufficiency and antisymmetry, which allows the use of modern sparse estimators.
- High empirical power when the proportion of nulls is high.

STRENGTHS IDENTIFIED INDEPENDENTLY

- The construction transfers the problem of calibrating a data-dependent statistic to a design-level exchangeability argument, so no distributional assumption on the statistic itself is needed. This

is the conceptual novelty and is why the method generalised later to arbitrary machine learning importance measures.

- The link to sequential testing places the filter in an established theory, makes the role of the plus correction transparent and provides a clean route to extensions.
- The paper is candid about the failure of the basic filter at $\rho = 0.9$ and ties it to a theoretical quantity, rather than hiding it.
- Code was released (R and Matlab packages), supporting replication.
- The orthogonal-design comparison with BHq identifies a mechanism (adaptivity to π_0) rather than just a performance number.

7.3 Weaknesses and critique

SINGLE MOST IMPORTANT WEAKNESS

Problem. The evidence on power rests on a narrow set of simulations that favour the method, against comparators that are either unguaranteed (BHq on least-squares z-scores) or deliberately weak (log-factor and whitening), and with no theory of power. **Why it matters.** The practical value of the method depends on whether the one-bit comparison loses little power relative to what is attainable, and the paper offers neither a bound nor a comparison with stronger valid competitors such as data splitting with Lasso screening or selective inference. **Potential bias.** The reported advantage may be overstated for realistic designs where signals are weaker, features are highly collinear (the equi-correlated s_j collapses) or n/p is close to one. **How it could be tested.** A systematic Monte Carlo over n/p, signal strength, correlation, and designs from real data, with Monte Carlo standard errors, comparing knockoffs with data splitting, selective inference and Gaussian-mirror style procedures.

OTHER WEAKNESSES

- Fixed-design and Gaussian assumptions: random designs, heteroskedasticity and serial dependence are excluded. Potential bias is inflated FDR in economic or time-series data. Test with block-bootstrapped or autocorrelated noise simulations.
- The requirement $n \geq p$ (and $n \geq 2p$ for the clean construction) excludes the genomics regime used to motivate the problem. The workaround with screening on a separate half halves the sample. Test the power loss against sample splitting.
- Randomness: the row-augmentation and the choice among equivalent orthonormal completions $\tilde{U}$ make the selected set depend on an auxiliary random draw. Different draws give different discoveries, which weakens reproducibility. Test by repeating the filter and measuring selection overlap.
- Threshold conservatism: the minimum number of discoveries at level q is $\lceil 1/q \rceil$ for knockoff+, and the procedure can return an empty set even with clear signals if few exist.
- Real-data validation without ground truth gives no information on FDR in practice. A semi-synthetic experiment that plants known effects in the real design would test it.

– Absence of standard errors in the simulation summaries means that small differences (for example SDP against equi-correlated) cannot be judged.

8 Contradictions and Edge Cases

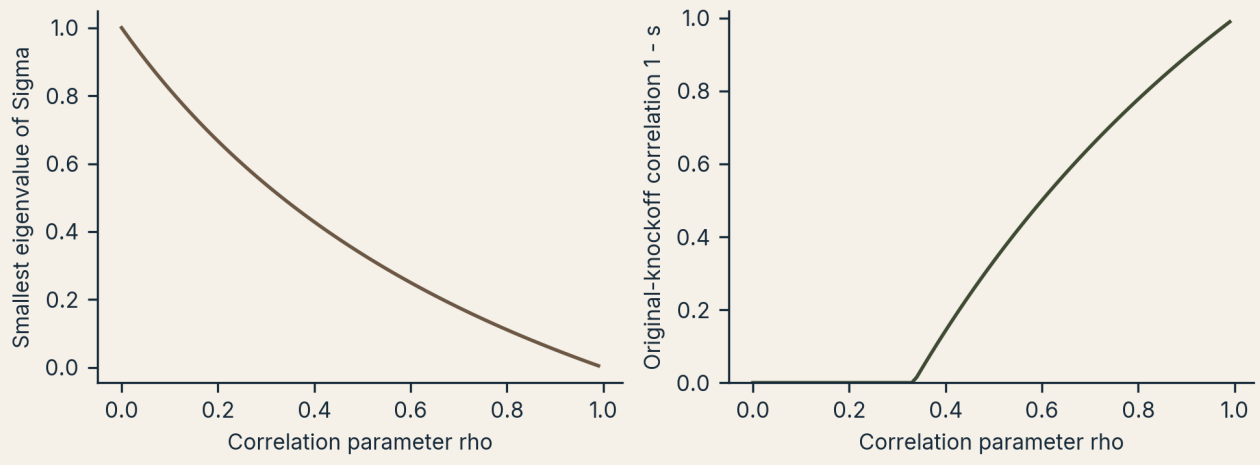
CRITIQUE Several conditions under which the result should weaken or disappear can be derived or computed directly.

Minimum number of discoveries. In knockoff+ the numerator is at least one, so a nonempty selection requires $1/R \leq q$, that is $R \geq \lceil 1/q \rceil$. At $q = 0.20$ at least 5 positive statistics are needed, at $q = 0.10$ at least 10, and at $q = 0.05$ at least 20. A study with 4 true strong signals cannot obtain any discovery at the 20 percent level with knockoff+, however strong they are. The basic filter has no such floor, which explains why it can select with a very small set but may violate FDR.

Basic filter under the global null. Theorem 1 bounds $\mathbb{E}[V/(R + q^{-1} - 1)]$. At $q = 0.2$ the denominator adds 4, so with $R = 1$ selection the bound allows $\mathbb{E}V/R$ as large as $5q = 1$ in the worst case. The reviewer simulation (Figure 2 of this review) confirms that the FDR of the basic filter under the global null is about 50 percent, while knockoff+ is about 3 to 4 percent. A practitioner who ignores this and reports the basic filter's output as FDR-controlled in a sparse problem is misled.

Equi-correlated knockoffs under strong correlation. For the AR(1) design $\Sigma_{jk} = \rho^{|j-k|}$ with large p, $\lambda_{\min}(\Sigma) \to (1-\rho)/(1+\rho)$. Then $s_j = 2(1-\rho)/(1+\rho)$ while this is below one, and $s_j = 1$ if and only if $\rho \leq 1/3$. The correlation between a variable and its knockoff is $1 - s_j$. The reviewer computed the exact smallest eigenvalue for $p = 1000$ (population Toeplitz matrix, Figure 3): $\lambda_{\min} = 0.333$ at $\rho = 0.5$, 0.111 at $\rho = 0.8$, 0.053 at $\rho = 0.9$, 0.026 at $\rho = 0.95$ and 0.005 at $\rho = 0.99$, which gives original-knockoff correlations of 0.333, 0.778, 0.895, 0.949 and 0.99. At $\rho = 0.9$ a knockoff is a near-copy of its original, so the pair cannot be distinguished by any statistic and power collapses. The paper's own failure at $\rho = 0.9$ matches the point at which this correlation exceeds 0.89. The SDP choice mitigates this only partially because it is constrained by the same condition $\mathrm{diag}\{s\} \preceq 2\Sigma$.

Figure 3. **Smallest eigenvalue of the AR(1) Gram matrix and equi-correlated original-knockoff correlation**



Source: generated by the reviewer. Population matrix with entries rho raised to the power |j-k|, p = 1000, rho from 0 to 0.99; s = min(2 times the smallest eigenvalue, 1). Sample Gram matrices with columns normalised approximate this for large n.

Heavy tails and sparse designs. Exchangeability needs exact Gaussian noise. With heavy-tailed noise, $X^\top z$ for sparse columns is non-Gaussian and the swap argument for the response fails, which the authors acknowledge is a hard case. The empirical excess of 0.31 points above nominal for knockoff+ suggests approximate robustness only. **Misspecification.** If the true mean contains an interaction or a nonlinear term, an included linear null feature correlated with it is not a null in the sense of (1), and the exchangeability argument no longer applies.

Alternative mechanism for the HIV agreement. Both BHq and knockoff select positions in the well-known panels because the strongest resistance mutations produce large OLS coefficients under almost any method. The agreement may therefore reflect the strength of the signals rather than the superiority of the FDR control, so the application cannot discriminate between procedures that are both reasonable.

Extreme values of q. As $q \to 0$, the threshold requires $R \geq 1/q$ positive signs with no negative sign, which will exceed p for $q < 1/p$ and force an empty selection. At the other end, $q \to 1$ gives the trivial rule that selects everything satisfying a ratio near one, with large power and unavoidable false discoveries.

9 Further Literature

Table 4. **Literature connections**

Paper	Shared idea	Relationship	Why it matters
Benjamini and Hochberg (1995)	Step-up control of the false discovery rate	Preceded and is the benchmark	Exactness under independence; the knockoff filter generalises the target to arbitrary design dependence.
Benjamini and Yekutieli (2001)	FDR under arbitrary dependence via a logarithmic factor	Preceded; shown by the authors to be very conservative	Provides the only prior guarantee for general dependence, and its power cost is what the knockoff paper avoids.
Storey (2002)	Estimation of the null proportion and of the FDR	Preceded	The adaptivity of knockoffs to π_0 in the orthogonal design reproduces the aim of Storey's estimator.
Tibshirani (1996)	Lasso as selection statistic	Foundation	Supplies the path and entry time Z_j from which W_j is built.
Meinshausen and Buhlmann (2010)	Stability selection with error control	Competing approach	Gives asymptotic selection guarantees and requires many refits; no general finite-sample FDR bound.
G'Sell, Wager, Chouldechova and Tibshirani (2016)	Sequential testing on the Lasso path	Developed a similar idea	Controls FDR only if all signals enter before all nulls, an assumption knockoffs avoid.

Paper	Shared idea	Relationship	Why it matters
Candes, Fan, Janson and Lv (2018)	Model-X knockoffs for random designs and any p	Extended	Removes the fixed-design, Gaussian-noise and $n \geq p$ requirements by sampling knockoffs from the feature distribution.
Barber and Candes (2019)	Knockoff filter for high-dimensional selective inference	Extended	Handles $p > n$ through a screening and knockoff split, addressing the main limitation noted in the discussion.
Barber, Candes and Samworth (2020)	Robustness of model-X knockoffs to estimated feature distributions	Improved the theory	Bounds FDR inflation when the knockoff generator is misspecified.
Weinstein, Barber and Candes (2017)	Power and prediction analysis of Lasso-based knockoffs	Filled the power gap	Gives an asymptotic power analysis, the missing piece of the original paper.
Fithian and Lei (2022)	Conditional calibration for FDR control	Alternative methodology	Offers a different route to FDR control under dependence for comparison.
Romano, Sesia and Candes (2020)	Deep knockoffs	Applied and extended	Generates knockoffs with neural networks for complex feature distributions.
Lee, Sun, Sun and Taylor (2016)	Exact post-selection inference	Contradicting framework	Controls inference conditional on selection rather than the FDR of the discovery list; a rival notion of validity.

Source: reviewer's construction. Bibliographic details of papers other than Barber and Candes (2015) are given from the reviewer's recollection and were not re-verified online in this run.

10 Research Gaps

10.1 Paper-level gaps

- **Gap:** No power theory. *Why it matters:* Users cannot predict when the one-bit comparison will lose power. *Existing limitation:* Power is shown by simulation only. *Way forward:* Asymptotic power of Lasso knockoff statistics, later started by Weinstein, Barber and Candes (2017), extended to finite samples.
- **Gap:** Behaviour with heavy tails and dependent errors. *Why it matters:* Applied data rarely have Gaussian i.i.d. noise. *Existing limitation:* One experiment. *Way forward:* Theorems on approximate exchangeability with bounds on FDR inflation.

10.2 Literature-level gaps

- **Gap:** Valid selection in time-series regression with persistent regressors. *Why it matters:* Finance and macroeconomics lack finite-sample error control in predictor selection. *Existing limitation:* Exchangeability arguments need independent noise. *Way forward:* Block knockoffs or blockwise swaps with dependent errors.
- **Gap:** Choosing among FDR-valid procedures. *Why it matters:* The literature offers knockoffs, data splitting, selective inference and e-value methods without a common benchmark. *Existing limitation:* Each paper uses its own simulations. *Way forward:* A public benchmark with fixed designs from real data.

10.3 Methodological gaps

- **Gap:** Reducing randomness. *Why it matters:* Different draws of $\tilde{U}$ yield different selected sets. *Existing limitation:* The paper constructs one knockoff. *Way forward:* Multiple knockoffs, derandomisation by aggregation, and e-value constructions.
- **Gap:** Scalable construction. *Why it matters:* The SDP and the Lasso on $2p$ variables become costly for p in the tens of thousands. *Existing limitation:* Solved only by the approximation of s. *Way forward:* Block-diagonal approximations of Σ and coordinate-wise solvers.

11 Research Opportunities

Idea 1. FDR-controlled selection of asset-pricing factors

RESEARCH QUESTION. Among a large set of published candidate factors, which ones carry independent explanatory power for a cross-section of returns once the false discovery rate is controlled under correlated factors?

MOTIVATION. The factor zoo contains hundreds of correlated candidates. Existing multiple-testing corrections use independent or bootstrapped t-statistics and do not exploit the joint design.

HYPOTHESIS. After knockoff-based FDR control at 10 percent, the number of independent factors is a small fraction of the zoo and concentrates in a few clusters.

ECONOMIC MECHANISM. Redundant factors proxy the same underlying risk premia, so their coefficients in a joint regression are mostly null.

LITERATURE CONNECTION. Harvey, Liu and Zhu (2016) on multiple testing in asset pricing; Feng, Giglio and Xiu (2020) on factor selection; Barber and Candes (2015).

DATA. Monthly long-short factor returns from the open-source asset pricing data set of Chen and Zimmermann and the Fama and French data library, both public; test portfolios from the Fama and French library.

VARIABLES. Response: excess return of a test portfolio or a time-series of the market-neutral return; regressors: candidate factors.

METHOD. Fixed-X knockoff+ with the row-augmentation extension for $p < n < 2p$, plus block-resampled calibration of the dependent errors.

IDENTIFICATION STRATEGY. Selection of the support of a linear pricing relation; no causal claim. Error control is checked by placebo factors (shuffled in time blocks).

EXPECTED CONTRIBUTION. A multiple-testing-valid list of factors under dependence among factors, with a comparison to the Benjamini and Hochberg style t-hurdle.

MAIN RISK. Return noise is heteroskedastic and serially dependent, which violates the Gaussian i.i.d. condition; the guarantee would be only approximate.

Idea 2. A reliability map of the knockoff filter

RESEARCH QUESTION. As a function of n/p, feature correlation, signal sparsity, noise kurtosis and design sparsity, where does the realised FDR of knockoff and knockoff+ depart from the nominal level, and by how much?

MOTIVATION. The paper tests one axis at a time and without Monte Carlo errors.

HYPOTHESIS. FDR inflation is largest when few discoveries are made and when the equi-correlated $1-s_j$ is close to one.

ECONOMIC MECHANISM. Not applicable; the mechanism is statistical (discreteness of one-bit p-values and the cost of near-collinear twins).

LITERATURE CONNECTION. Barber and Candes (2015), Barber, Candes and Samworth (2020), Weinstein, Barber and Candes (2017).

DATA. Simulated designs, and designs drawn from public data sets such as the FRED-MD macro panel.

VARIABLES. Realised FDR, power, number of discoveries, $\lambda_{\min}(\Sigma)$, n/p, excess kurtosis.

METHOD. Large Monte Carlo experiment with standard errors, followed by a regression of FDR inflation on design diagnostics.

IDENTIFICATION STRATEGY. Controlled experimental variation of each factor of the design.

EXPECTED CONTRIBUTION. A practical diagnostic that tells a user when to trust the nominal level.

MAIN RISK. Results may be specific to the simulated families.

Idea 3. Selecting macro predictors for nowcasting under FDR control

RESEARCH QUESTION. Which indicators in a large macro panel have incremental predictive content for a target such as industrial production growth, with FDR control?

MOTIVATION. Large panels are used with ad hoc selection (Lasso with cross-validation), leaving the number of spurious predictors unknown.

HYPOTHESIS. Only a small number of indicators survive FDR control at 10 percent, and the selected set is less stable across vintages than the Lasso set.

ECONOMIC MECHANISM. Common factors loading on many indicators make the individual indicators nearly redundant.

LITERATURE CONNECTION. Stock and Watson on diffusion indexes; McCracken and Ng (2016) on FRED-MD; Barber and Candes (2015).

DATA. FRED-MD monthly data, about 128 series and roughly 700 to 800 monthly observations, public.

VARIABLES. Response: growth of industrial production h months ahead; regressors: lagged transformed indicators.

METHOD. Knockoff+ with row-augmentation (with p about 128 and n about 700 to 800 the plain $n \geq 2p$ construction applies), with a stationary block-bootstrap of the null W signs as a check.

IDENTIFICATION STRATEGY. Prediction, not causation; out-of-sample evaluation by expanding windows.

EXPECTED CONTRIBUTION. Evidence on the stability of nowcasting predictors with error control.

MAIN RISK. Autocorrelated residuals violate exchangeability.

Idea 4. Multiple knockoffs and derandomisation

RESEARCH QUESTION. How much power does the one-bit comparison lose, and can averaging several independent knockoff draws recover it?

MOTIVATION. The authors identify extreme discretisation as a source of conservatism at low amplitude.

HYPOTHESIS. Aggregating m draws raises power at low amplitude while keeping FDR near nominal.

ECONOMIC MECHANISM. Not applicable; information loss from one bit per variable.

LITERATURE CONNECTION. Barber and Candes (2015) Section 6; later multi-knockoff and e-value work.

DATA. Simulated designs and the HIV data of Rhee et al. (2006), which is public.

VARIABLES. Power, FDR, selection overlap between draws.

METHOD. Repeat the filter with $m = 1, 5, 20$ draws and aggregate by selection frequency or by e-value combination.

IDENTIFICATION STRATEGY. Controlled simulation; semi-synthetic planting in the real HIV design.

EXPECTED CONTRIBUTION. Quantified power-reproducibility trade-off.

MAIN RISK. Aggregation may break the exact guarantee.

Idea 5. FDR-controlled predictors of realised volatility

RESEARCH QUESTION. Which lagged and cross-asset variables add predictive power for next-period volatility beyond the heterogeneous autoregressive model, with error control?

MOTIVATION. Lagged volatility measures at many horizons are highly correlated, the regime where selection is delicate.

HYPOTHESIS. After FDR control, only the daily, weekly and monthly components plus a small number of implied-volatility terms remain.

ECONOMIC MECHANISM. Heterogeneous trading horizons create persistence at several frequencies (Corsi, 2009), which are correlated but not redundant.

LITERATURE CONNECTION. Corsi (2009) HAR model; Barber and Candes (2015); Barber and Candes (2019) for $p > n$.

DATA. Daily S&P 500 and volatility index series from FRED or other public sources, squared returns as a volatility proxy.

VARIABLES. Next-day log squared return; regressors: lagged volatility terms at several horizons, volatility index, credit spread.

METHOD. Knockoff+ with SDP knockoffs, with heteroskedasticity-robust standardisation of the response.

IDENTIFICATION STRATEGY. Predictive; out-of-sample forecast comparison.

EXPECTED CONTRIBUTION. Parsimonious volatility model with a known error rate on included terms.

MAIN RISK. Volatility proxy is noisy and heavy-tailed; the equi-correlated s_j collapses for highly collinear lags.

11.1 Ranking

Table 5. **Ranking of research ideas (scores from 1 to 5, assigned by the reviewer)**

Idea	Novelty	Feasibility	Data	Method	Importance	Contribution	Total
1. Asset-pricing factors	4	4	5	3	5	4	25
2. Reliability map	3	5	5	4	4	4	25
3. Macro nowcasting	3	4	5	3	4	3	22
4. Multiple knockoffs	3	4	4	4	3	3	21
5. Volatility predictors	4	3	4	3	4	3	21

Source: reviewer's subjective scoring.

The top three are ideas 1, 2 and 3. Idea 1 addresses a question of first-order importance in empirical finance with fully public data, and the dependence in returns is a well-posed methodological challenge. Idea 2 is the most feasible and is a prerequisite for the others, since it measures how far the nominal level can be trusted. Idea 3 combines public data with a regime ($p < n$) in which the fixed-design theory applies directly.

12 Learning Framework

12.1 Prerequisite concepts

- Linear regression in matrix form, the Gram matrix and its eigenvalues.
- The Lasso and its solution path; the notion of entry time.
- Multiple testing, the family-wise error rate, the false discovery rate and the Benjamini and Hochberg step-up procedure.
- Positive semidefinite matrices, Schur complements and semidefinite programming.
- Stopping times, supermartingales and optional stopping.

12.2 Core concepts learned

- Exchangeability of null variables with their knockoffs gives fair-coin signs, which converts an unknown null distribution into a known one.
- A ratio of negative to positive statistics is a conservative estimate of the false discovery proportion.
- Adding one to the numerator (the plus) turns approximate into exact control.
- Permutation copies are not knockoffs because they destroy correlation with other originals.

12.3 Important equations

The Gram conditions (3) define a knockoff, the thresholds (10) define the filters, inequality (11) explains why the ratio estimates the false discovery proportion and the identity (13) explains why the plus yields a bound of one. The choice (6) shows why a small eigenvalue of Σ limits power.

12.4 Methods learned

Construction of a design-level surrogate with prescribed second-order structure, use of antisymmetric statistics, stopping-time arguments for data-dependent thresholds, and reduction of a regression problem to sequential testing with one-bit p-values.

12.5 Questions to test understanding

COMPREHENSION

1. Why must $X^\top \tilde{X}$ equal $\Sigma - \mathrm{diag}\{s\}$ rather than Σ?
2. What does a negative W_j for a non-null variable mean, and why does it not invalidate the estimator?
3. Why is the basic knockoff filter not an FDR-controlling procedure?

ANALYSIS

1. Prove that $G \succeq 0$ iff $\mathrm{diag}\{s\} \succeq 0$ and $2\Sigma \succeq \mathrm{diag}\{s\}$.
2. Show that if signals always enter before knockoffs and nulls are fair coins, the ratio in (10) is an unbiased-or-conservative estimate of the proportion.
3. Why does the log-factor correction of Benjamini and Yekutieli cost so much power in the simulation?

RESEARCH

1. How would the exchangeability argument change if the noise were autocorrelated with known covariance?
2. Can a single knockoff be improved by choosing s adaptively using y without breaking Lemma 1?
3. What is the best achievable power for a one-bit rule, and how far is the Lasso statistic from it?

12.6 Papers to read next

Candes, Fan, Janson and Lv (2018); Barber and Candes (2019); Weinstein, Barber and Candes (2017); Benjamini and Yekutieli (2001); G'Sell, Wager, Chouldechova and Tibshirani (2016).

13 Conclusion

13.1 One-sentence thesis

Constructing synthetic variables that are exchangeable with the nulls turns the unknown null distribution of a selection statistic into a fair coin and thereby gives exact finite-sample false discovery rate control for regression with arbitrary correlated designs.

13.2 Three most important insights

- Calibration can be obtained from the design, not from the response: the knockoffs are built from X alone, so the guarantee does not depend on the noise level or the signal.
- The plus correction is not cosmetic: the reviewer simulation gives a global-null FDR of about 50 percent for the basic filter against about 3 to 4 percent for knockoff+.
- The power of the method is limited by how distinguishable a variable is from its knockoff, which is governed by the smallest eigenvalue of the Gram matrix through equation (6).

13.3 Strongest evidence

The proof of exact control (Lemma 1, the bound (12) and the binomial identity (13)), together with Table 1 where knockoff+ attains an FDR near 14 to 15 percent and power of about 61 percent against 19 and 2 percent for the valid Benjamini and Hochberg variants.

13.4 Biggest weakness

Power is supported by narrow simulations against weak comparators and has no theory.

13.5 Most important research gap

Valid and powerful selection when errors are dependent or non-Gaussian and when p exceeds n.

13.6 Best research opportunity

Apply the filter to factor selection in empirical asset pricing with dependence-robust calibration (Idea 1), preceded by a map of where the nominal level fails (Idea 2).

13.7 Three papers to read next

Candes, Fan, Janson and Lv (2018); Barber and Candes (2019); Benjamini and Yekutieli (2001).

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